## Analysis of Building Frames (Using Kani's Method)

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.

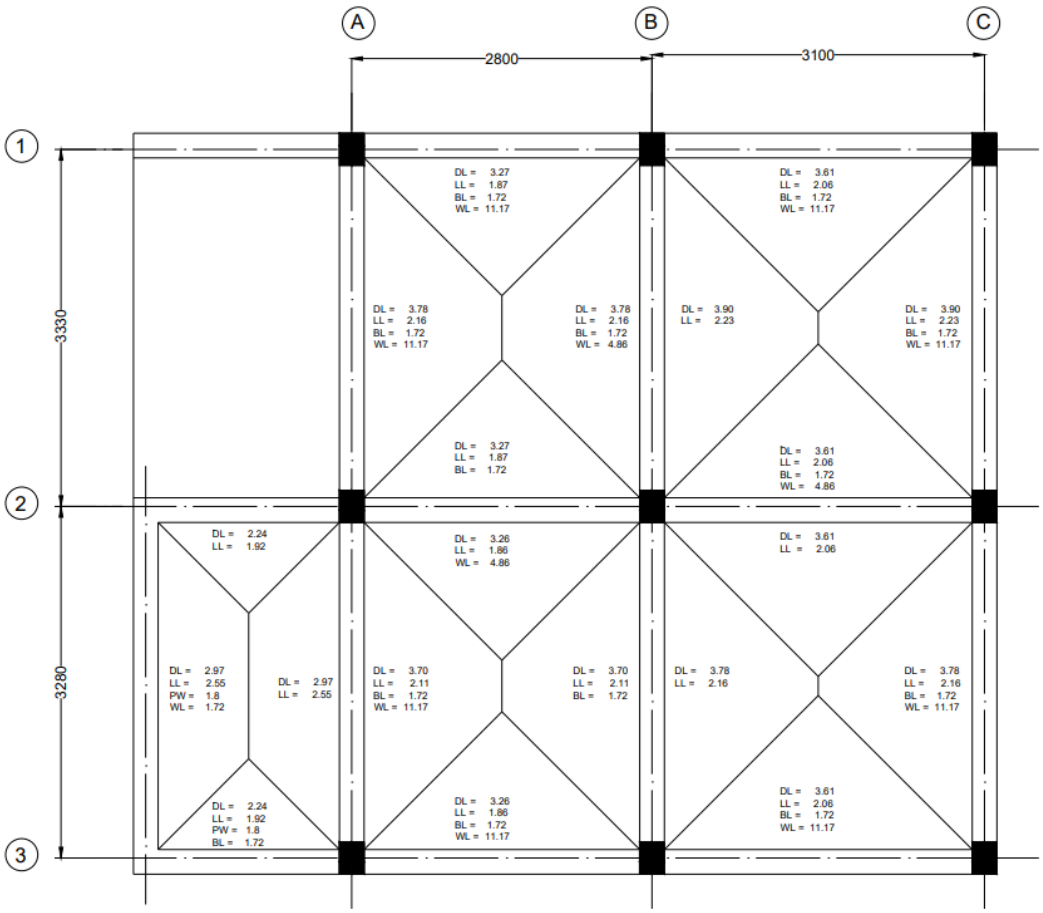



***Figure 1: Beam Layout***

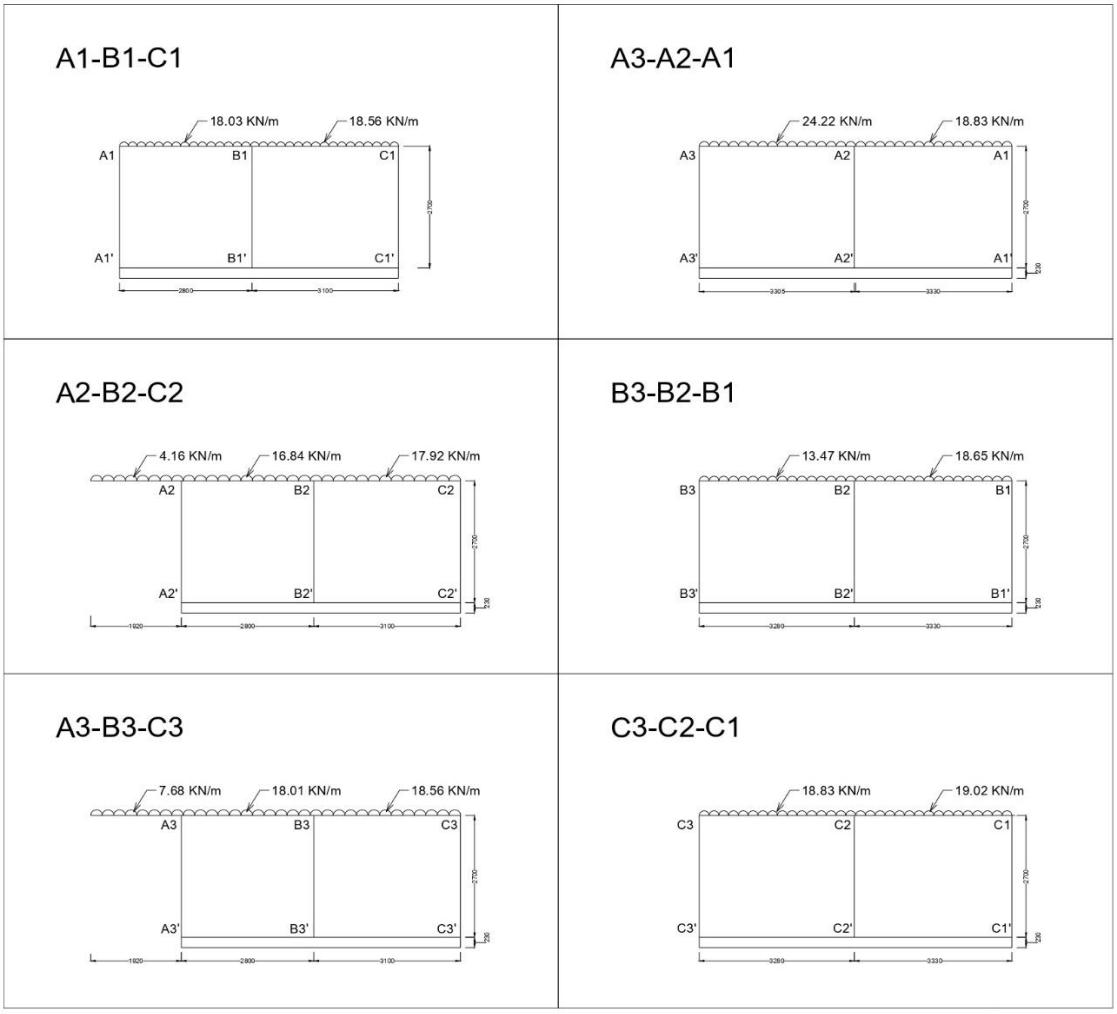
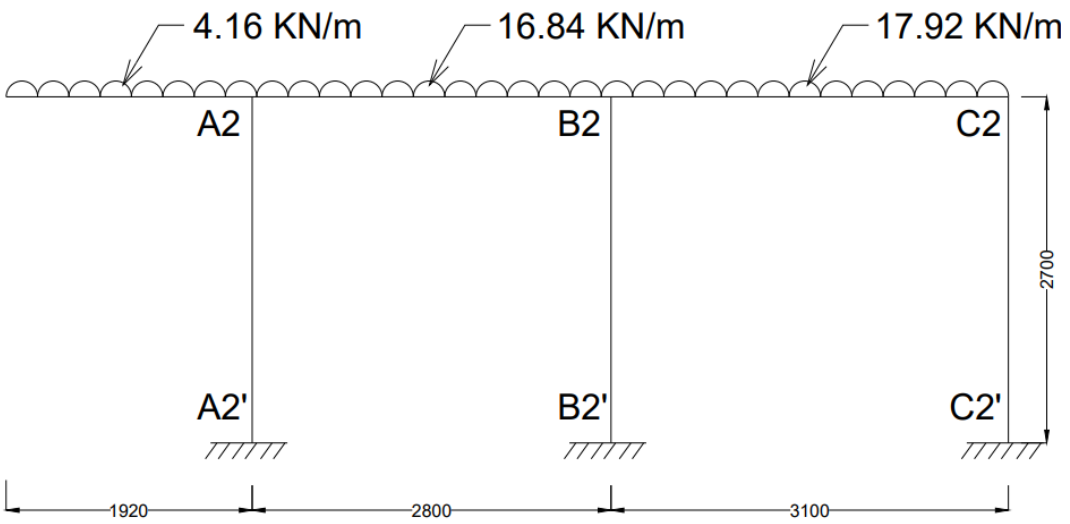



***Figure 2:*** Frames.

**Analysis of Frame A2-B2-C2**




**Step 1: Fixed End Moments**

$MF_{Overhanging} = \frac{WL^2}{2} = \frac{4.16 \times 1.92^2}{2} = 7.67$ KN-m

$MF_{A2B2} = \frac{-WL^2}{12} = \frac{-16.84 \times 2.8^2}{12} = -11.00$ KN-m

$MF_{B2A2} = \frac{WL^2}{12} = \frac{-16.84 \times 2.8^2}{12} = +11.00$ KN-m

$MF_{B2C2} = \frac{-WL^2}{12} = \frac{-17.92 \times 3.1^2}{12} = -14.35$ KN-m

$MF_{C2B2} = \frac{WL^2}{12} = \frac{17.92 \times 3.1^2}{12} = +14.35$ KN-m

Fixed end moments in All the columns are zero.

i.e,

$MF_{A2A2'} = MF_{A2'A2} = MF_{B2B2'} = MF_{B2'B2} = MF_{C2C2'} = MF_{C2'C2} = 0$

**Step 2: Rotation Factor** $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right), \left(k = \frac{4EI}{L}\right)$

| Joints | Members | Stiffness ($k$) | $\Sigma k$ | R.F |
|---|---|---|---|---|
| A2 | A2B2 | 1.428EI | 2.909EI | -0.245 |
| | A2A2′ | 1.481EI | | -0.255 |
| B2 | B2A2 | 1.428EI | 4.200EI | -0.170 |
| | B2C2 | 1.290EI | | -0.154 |
| | B2B2′ | 1.481EI | | -0.176 |
| C2 | C2B2 | 1.290EI | 2.771EI | -0.233 |
| | C2C2′ | 1.481EI | | -0.267 |

**Stiffness ($k$)** $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A2B2} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{A2A2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B2A2} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{B2C2} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{B2B2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C2B2} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{C2C2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

**Rotation Factor** $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$R.F_{A2B2} = -\frac{1}{2}\left(\frac{1.428EI}{2.909EI}\right) = -0.245$

$R.F_{A2A2'} = -\frac{1}{2}\left(\frac{1.481EI}{2.909EI}\right) = -0.255$

$R.F_{B2A2} = -\frac{1}{2}\left(\frac{1.428EI}{4.200EI}\right) = -0.170$

$R.F_{B2C2} = -\frac{1}{2}\left(\frac{1.290EI}{4.200EI}\right) = -0.154$

$R.F_{B2B2'} = -\frac{1}{2}\left(\frac{1.481EI}{4.200EI}\right) = -0.176$

$R.F_{C2B2} = -\frac{1}{2}\left(\frac{1.290EI}{2.771EI}\right) = -0.233$

$R.F_{C2C2'} = -\frac{1}{2}\left(\frac{1.481EI}{2.771EI}\right) = -0.267$

**Sum of Stiffness $\sum k$ at Joints**

A2: 1.428EI + 1.481EI = 2.909EI

B2: 1.428EI + 1.290EI + 1.481EI = 4.200EI

C2: 1.290EI + 1.481EI = 2.771EI

**Step 3: Displacement Factor** $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

| Storey | Members | $k$ | $\Sigma k$ | D.F |
|---|---|---|---|---|
| 1 | A2A2′ | 1.481EI | 4.444EI | -0.5 |
| | B2B2′ | 1.481EI | | -0.5 |
| | C2C2′ | 1.481EI | | -0.5 |

**Stiffness ($k$)** $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A2A2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B2B2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C2C2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

**Displacement Factor** $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$D.F_{A2A2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{B2B2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{C2C2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

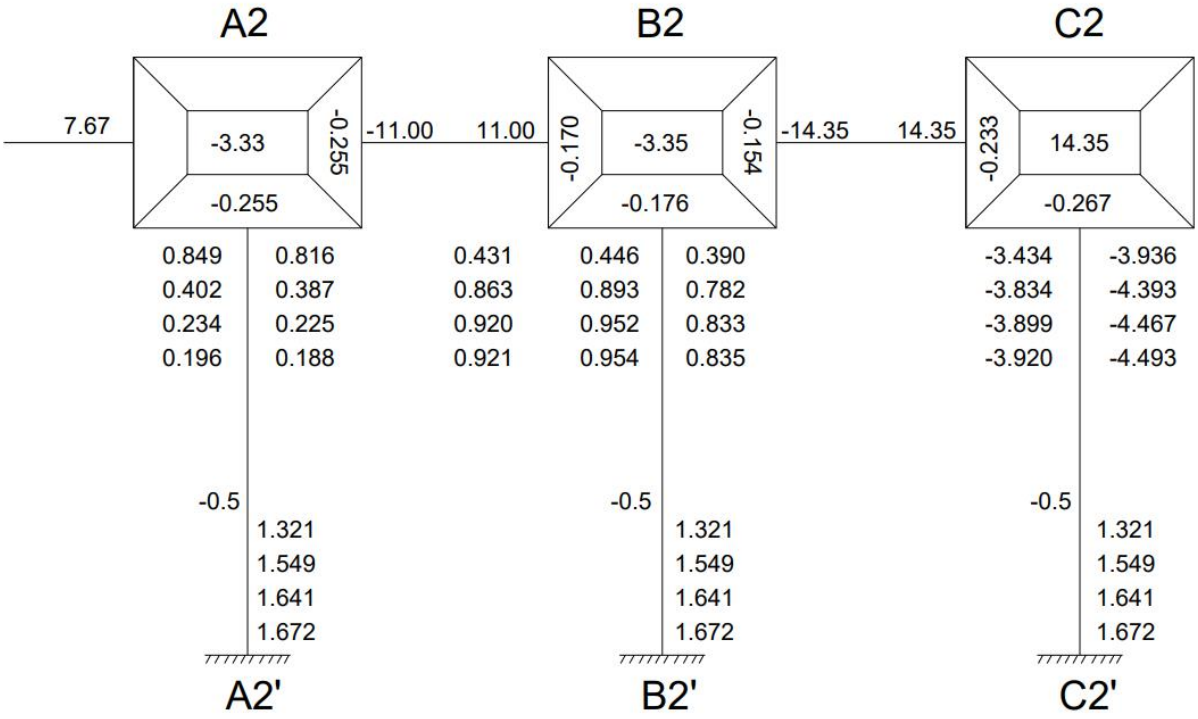
**Sum of Stiffness $\sum k$ at Storey**

$\sum k$ = 1.481EI + 1.481EI + 1.481EI = 4.444EI

## Rotation and Displacement Contribution

starting FEM at Joint A2 ∑FEM will be the sum of the overhang and the beam:

$\sum FEM_{A2}$ = 7.67 + (-11.00) = -3.33KN-m



**Cycle 1**

**Joint Rotation** $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

$M_{A2B2}$ = -0.245 × [-3.33 + 0 + 0] = 0.816

$M_{A2A2'}$ = -0.255 × [-3.33 + 0 + 0] = 0.849

$M_{B2A2}$ = -0.170 × [-3.35 + 0.816 + 0] = 0.431

$M_{B2C2}$ = -0.154 × [-3.35 + 0.816 + 0] = 0.390

$M_{B2B2'}$ = -0.176 × [-3.35 + 0.816 + 0] = 0.446

$M_{C2B2}$ = -0.233 × [14.35 + 0.390 + 0] = -3.434

$M_{C2C2'}$ = -0.267 × [14.35 + 0.390 + 0] = -3.936

**Displacement Contribution**

**For All Columns** $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'}$ = -0.5 × (0.849 + 0.446 + (-3.936)) = 1.321

**Cycle 2**

**Joint Rotation** $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

$M_{A2B2}$ = -0.245 × [-3.33 + 1.321 + 0.431] = 0.387

$M_{A2A2'}$ = -0.255 × [-3.33 + 1.321 + 0.431] = 0.402

$M_{B2A2}$ = -0.170 × [-3.35 + 1.321 + 0.387 + (-3.434)] = 0.863

$M_{B2C2}$ = -0.154 × [-3.35 + 1.321 + 0.387 + (-3.434)] = 0.782

$M_{B2B2'}$ = -0.176 × [-3.35 + 1.321 + 0.387 + (-3.434)] = 0.893

$M_{C2B2}$ = -0.233 × [14.35 + 1.321 + 0.782] = -3.834

$M_{C2C2'}$ = -0.267 × [14.35 + 1.321 + 0.782] = -4.393

**Displacement Contribution**

**For All Columns** $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'}$ = -0.5 × (0.402 + 0.893 + (-4.393)) = 1.549

**Cycle 3**

**Joint Rotation** $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

$M_{A2B2}$ = -0.245 × [-3.33 + 1.549 + 0.863] = 0.225

$M_{A2A2'}$ = -0.255 × [-3.33 + 1.549 + 0.863] = 0.234

$M_{B2A2}$ = -0.170 × [-3.35 + 1.549 + 0.225 + (-3.834)] = 0.920

$M_{B2C2}$ = -0.154 × [-3.35 + 1.549 + 0.225 + (-3.834)] = 0.833

$M_{B2B2'}$ = -0.176 × [-3.35 + 1.549 + 0.225 + (-3.834)] = 0.952

$M_{C2B2}$ = -0.233 × [14.35 + 1.549 + 0.833] = -3.899

$M_{C2C2'}$ = -0.267 × [14.35 + 1.549 + 0.833] = -4.467

**Displacement Contribution**

**For All Columns** $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'}$ = -0.5 × (0.234 + 0.952 + (-4.467)) = 1.641

**Cycle 4**

**Joint Rotation** $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

$M_{A2B2}$ = -0.245 × [-3.33 + 1.641 + 0.920] = 0.188

$M_{A2A2'}$ = -0.255 × [-3.33 + 1.641 + 0.920] = 0.196

$M_{B2A2}$ = -0.170 × [-3.35 + 1.641 + 0.188 + (-3.899)] = 0.921

$M_{B2C2}$ = -0.154 × [-3.35 + 1.641 + 0.188 + (-3.899)] = 0.835

$M_{B2B2'}$ = -0.176 × [-3.35 + 1.641 + 0.188 + (-3.899)] = 0.954

$M_{C2B2}$ = -0.233 × [14.35 + 1.641 + 0.835] = -3.920

$M_{C2C2'}$ = -0.267 × [14.35 + 1.641 + 0.835] = -4.493

**Displacement Contribution**

**For All Columns** $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A2A2'} + M_{B2B2'} + M_{C2C2'}$ = -0.5 × (0.196 + 0.954 + (-4.493)) = 1.672

**Step 4: Final End Moments**

**$M_{ij}$ = FEM + 2×$M_{ij}$ (Near end Rotation) + $M_{ji}$ (Far end Rotation) +M″ (Displacement)**

**Beams (No Sway M″= 0) "for Joint A2, the Overhang Moment is a fixed constant of +7.67 KN-m"**

$M_{A2B2}$ = -11.00 + 2(0.188) + 0.921 = -9.70

$M_{B2A2}$ = 11.00 + 2(0.921) + 0.188 = 13.03

$M_{B2C2}$ = -14.35+ 2(0.835) + (-3.920) = -16.60

$M_{C2B2}$ = 14.35 + 2(-3.920) + 0.835 = 7.35

**Column (With Sway M″= 1.672)**

$M_{A2A2'}$ = 0 + 2(0.196) + 0 + 1.672 = 2.06

$M_{A2'A2}$ = 0 + 2(0) + 0.196 + 1.672 = 1.87

$M_{B2B2'}$ = 0 + 2(0.954) + 0 + 1.672 = 3.58

$M_{B2'B2}$ = 0 + 2(0) + 0.954 + 1.672 = 2.63

$M_{C2C2'}$ = 0 + 2(-4.493) + 0 + 1.672 = -7.31

$M_{C2'C2}$ = 0 + 2(0) + (-4.493) + 1.672 = -2.82

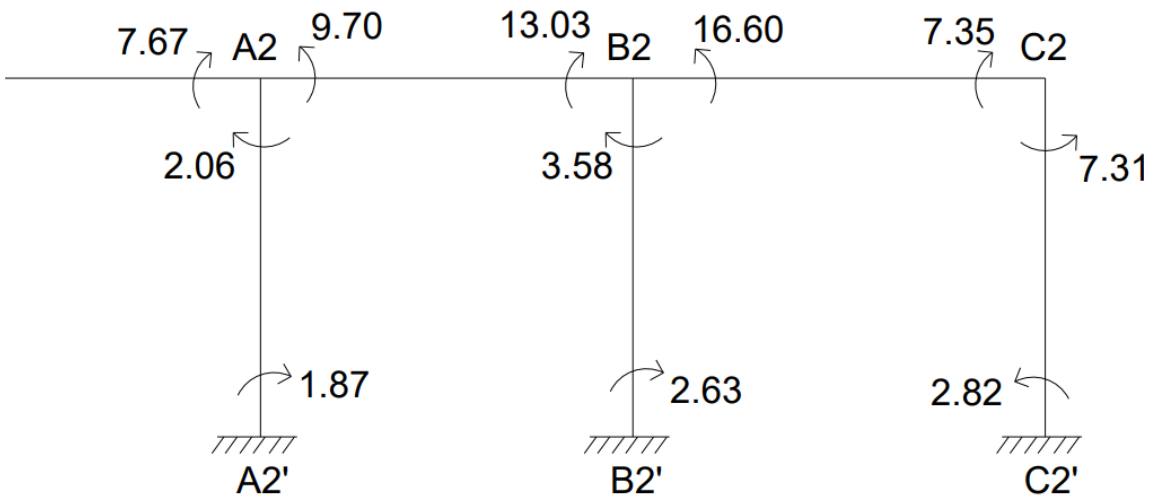
**Equilibrium Check**

"To check Joint A2, we must include that +7.67 overhanging moment along with the beam and column."

Joint A2 = $M_{overhanging}$ + $M_{A2B2}$ + $M_{A2A2'}$ = 7.67 + (-9.70) + 2.06 = 0.03 ≈ 0

Joint B2 = $M_{B2A2}$ + $M_{B2C2'}$ + $M_{B2B2'}$ = 13.03 + (-16.60) + 3.58 = 0.01 ≈ 0

Joint C2 = $M_{C2B2}$ + $M_{C2C2'}$ = 7.35 – 7.31 = 0.04 ≈ 0



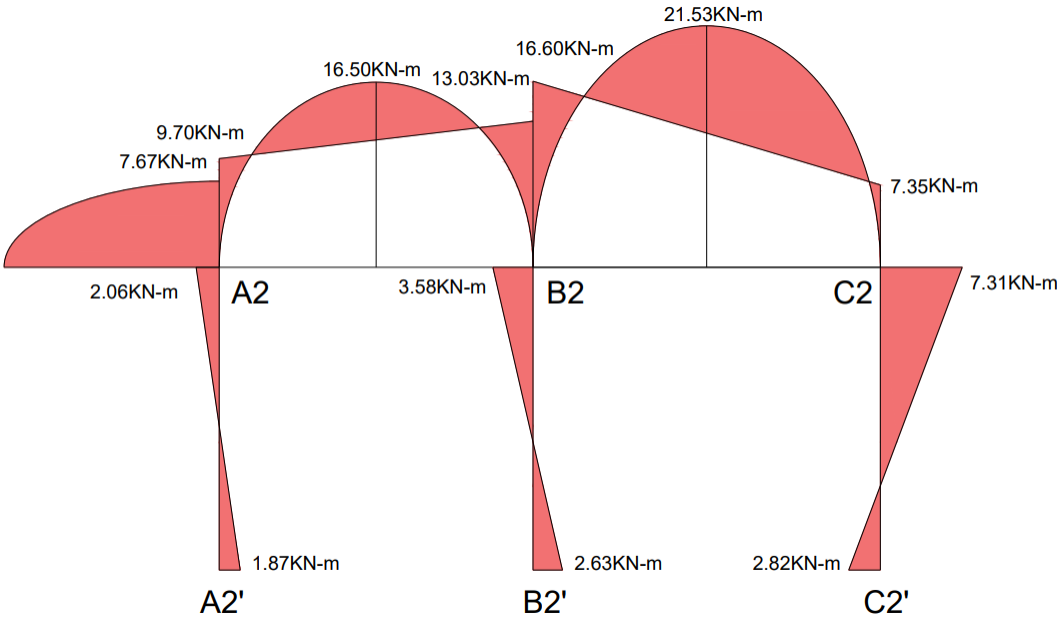
## CALCULATION BENDING MOMENT

"Since the overhang is a cantilever, it doesn't have a simply supported "sagging" mid-span moment. Its moment is purely negative (hogging), starting at 0 at the free end and curving to 7.67 KN-m at support A2."

$BM_{A2B2} = \frac{WL^2}{8} = \frac{16.84 \times 2.8^2}{8} = 16.50KN\text{-}m$

$BM_{B2C2} = \frac{WL^2}{8} = \frac{17.92 \times 3.1^2}{8} = 21.53KN\text{-}m$

## BENDING MOMENT DIAGRAM




## CALCULATION OF REACTIONS

From Overhang (Direct load transfer):

$R_{overhang} = w \times L = 4.16 \times 1.92 = 7.99KN$

### For Span A2B2

Simply Supported Reactions:

$R_{total}$ @ A2B2$= \frac{WL}{2} = \frac{16.84 \times 2.8}{2} = 23.58KN$

Reaction Due to End Moments: (Net clockwise moment means left support gets less load)

$R_{moment}$ @ A2B2$= \frac{M_{A2B2} - M_{B2A2}}{L} = \frac{-9.70 + 13.03}{2.8} = -1.19KN$

Reaction Due to End Moments: (Net clockwise moment means right support gets more load)

$R_{moment}$ @ A2B2$= \frac{M_{A2B2} - M_{B2A2}}{L} = \frac{+9.70 + 13.03}{2.8} = +1.19KN$

**For Span B2C2**

Simply Supported Reactions:

$R_{total}$ @ B2C2= $\frac{WL}{2} = \frac{17.92 \times 3.1}{2}$ = 27.78KN

Reaction Due to End Moments: (Net clockwise moment means right support gets less load)

$R_{moment}$ @ B2C2= $\frac{M_{B2C2} - M_{C2B2}}{L} = \frac{-16.60 + 7.35}{3.1}$ = -2.98KN

Reaction Due to End Moments: (Net clockwise moment means left support gets more load)

$R_{moment}$ @ B2C2= $\frac{M_{B2C2} - M_{C2B2}}{L} = \frac{16.60 + 7.35}{3.1}$ = +2.98KN

**Final Reactions:**

$R_{A2} = R_{overhang} + R_{total}$ @ A2B2+ $R_{moments}$ @ $A2B2_{(Left\ Support)}$

$R_{A2}$ =7.99 + 23.58 + (-1.19) = 30.38KN

$R_{B2} = (R_{total}$ @ A2B2+ $R_{moments}$ @ $A2B2_{(Right\ Support)}) + (R_{total}$ @ B2C2+ $R_{moments}$ @ $B2C2_{(Left\ Support)})$ =

$R_{B2}$ = (23.58 + 1.19) + (27.78 + 2.98) = 55.53

$R_{C2} = R_{total}$ @ B2C2+ $R_{moments}$ @ $B2C2_{(Right\ Support)}$ =

$R_{C2}$ = 27.78 + (-2.98) = 24.80 KN

